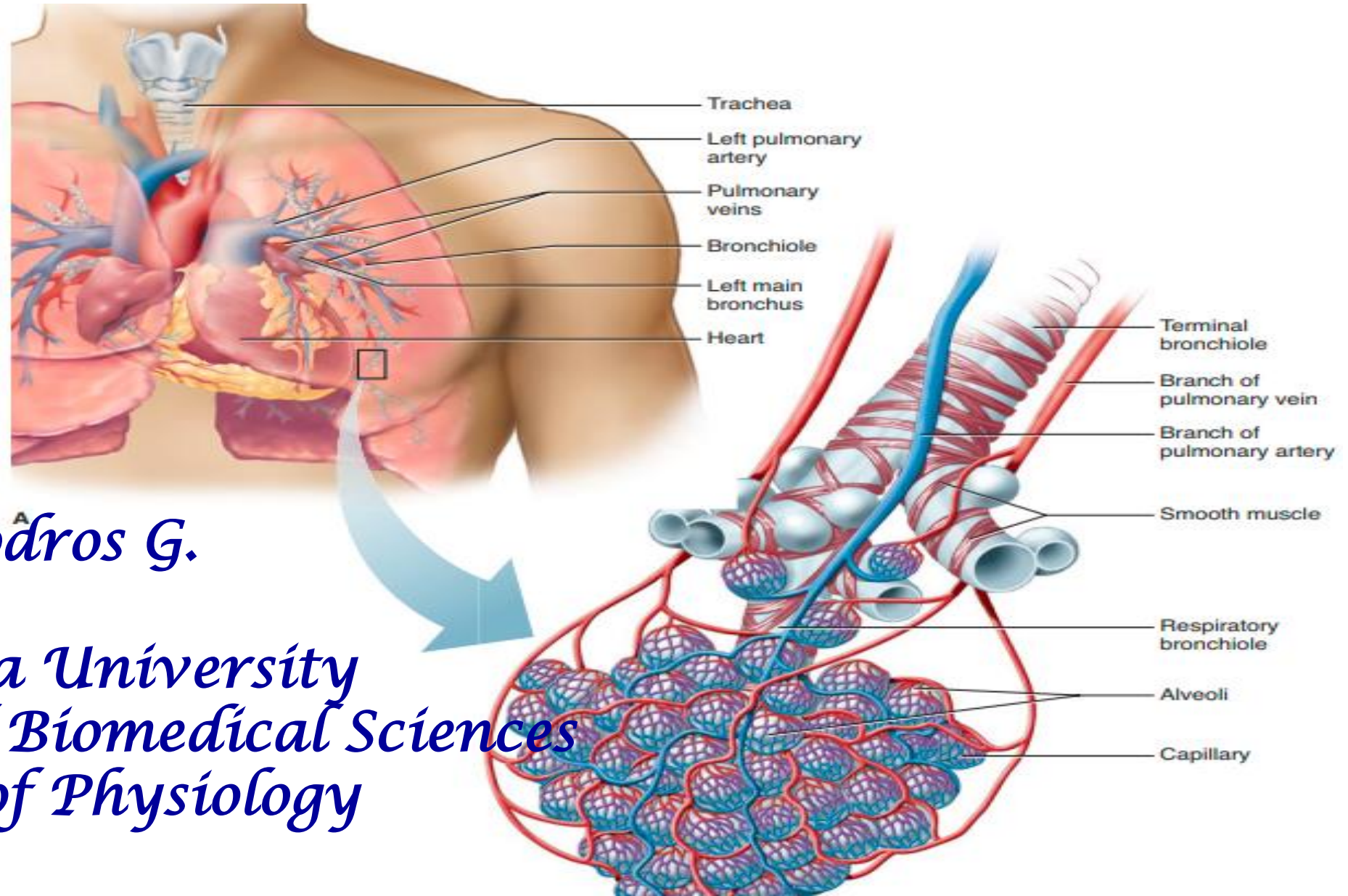


Physiology of Respiratory System



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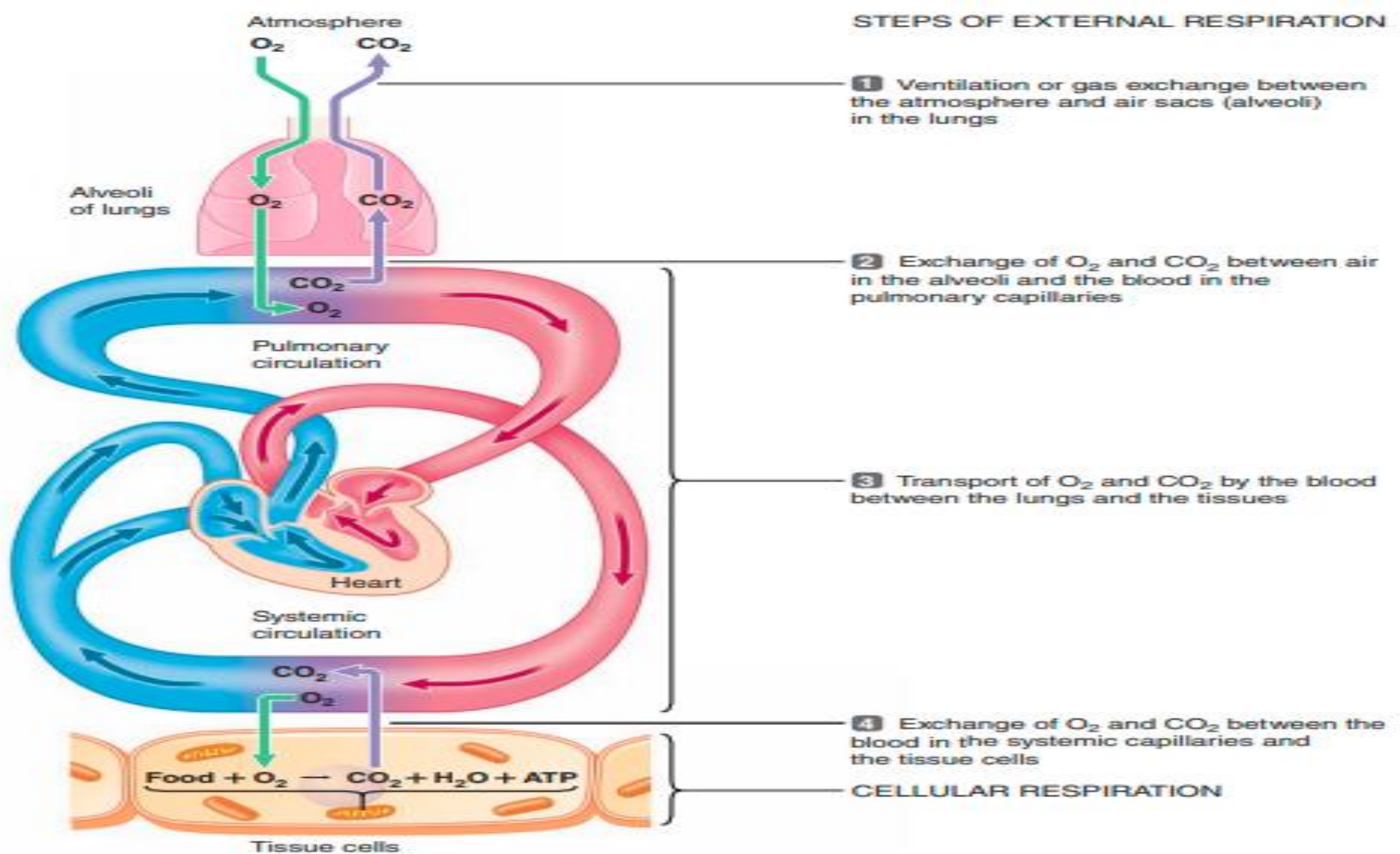
■ Respiration

- ❑ Actual exchange of gases in lungs and tissues
- ❑ **Respiratory and Circulatory Systems** work together to deliver O_2 to cells and remove CO_2 by the two-phase of **respiration**

☒ **The 1st phase of respiration** begins with **breathing in**

- **Inhalation** brings air from outside of the body into the lungs.
- Oxygen moves from the lungs through blood vessels to the heart, which pumps the **oxygen-rich blood** to all parts of the body.
- O_2 moves from the bloodstream into cells, which **completes the first phase of respiration**

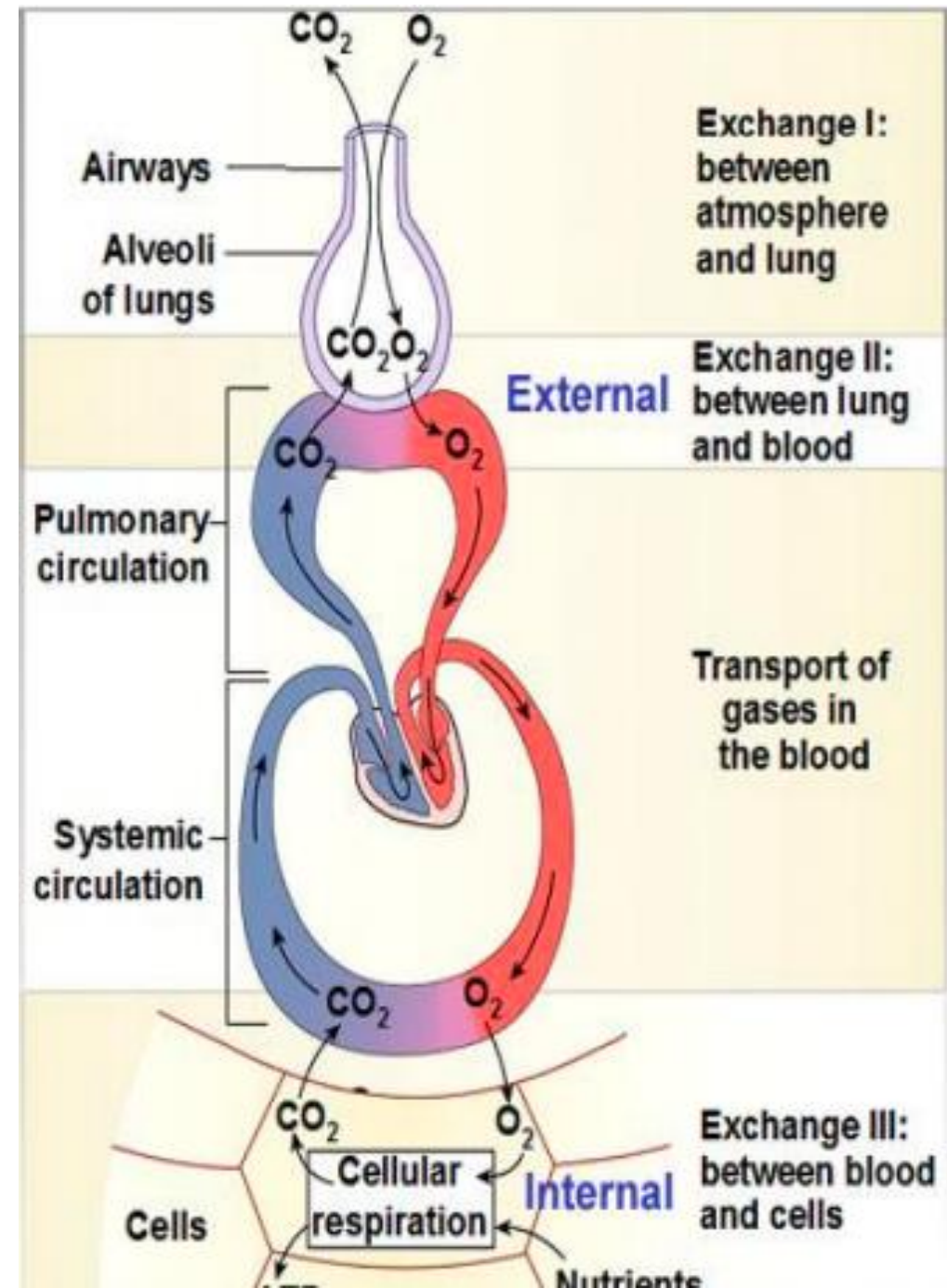
STEPS OF EXTERNAL RESPIRATION



- ❑ **In the Cells**, oxygen is used for energy-producing process called **cellular respiration**, which produces CO₂ as a by product.
- ☒ **The 2nd phase of respiration** begins with the movement of **CO₂** from the **cells** to the **bloodstream**.
 - Bloodstream carries CO₂ to the heart, which pumps the **carbon dioxide-loaden blood** to the **lungs**.
- ❑ **In the Lungs**, **exhalation** removes CO₂ from the body, thus completing the **Respiratory Cycle**

Phases of Respiration

- ❑ There are three phases of respiration.
 1. **Pulmonary ventilation**
 2. **External respiration**
 3. **Internal respiration**
- ☒ **Pulmonary ventilation** (breathing)
 - ⇒ b/n atmosphere and lungs
- ☒ **External respiration**
 - ⇒ b/n alveoli and pulmonary capillaries
- ☒ **Internal respiration** (Cellular respiration)
 - ⇒ b/n systemic capillaries and cells
- ❑ O_2 is utilized for metabolic activities
- ❑ CO_2 is produced as a result of metabolic activities



❑ General Functions of Respiratory System

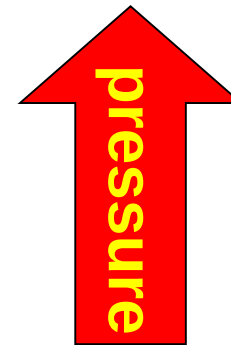
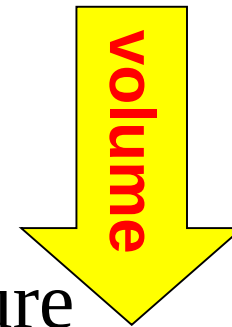
1. Gas exchange
2. Regulate the balance of acid and base in tissues
3. Protects the body against disease-causing organisms and toxic substances inhaled with air.
4. House of the cells that detect smell, and assists in the production of sounds for speech.
5. Angiotensin converting enzyme secretion (ACE).
6. Control of body temperature due to loss of evaporate during expiration

Mechanics of Breathing (Pulmonary Ventilation)

- ❑ **Pulmonary Ventilation** is the inflow and outflow of air between the atmosphere and the lungs
- ❑ The action of breathing in and out is due to **changes of pressure within the thorax and the lungs**, in comparison with the **atmosphere pressure**
- ❑ In accordance with *Boyle's Law*, when the volume of a given quantity of gas increases its pressure decreases and vice versa at constant temperature:

$$V \propto \frac{1}{p}$$

⇒ Volume is inversely proportional to pressure



❑ Air flows from **area of higher pressure** to area of lower pressure

☒ **Pulmonary ventilation** is brought by changing intrapulmonary pressure to subatmospheric pressure and above atmospheric pressure during **inspiration** and **expiration** respectively

❑ **Breathing (Pulmonary Ventilation)**

⇒ has two phases

1. Inspiration/Inhalation

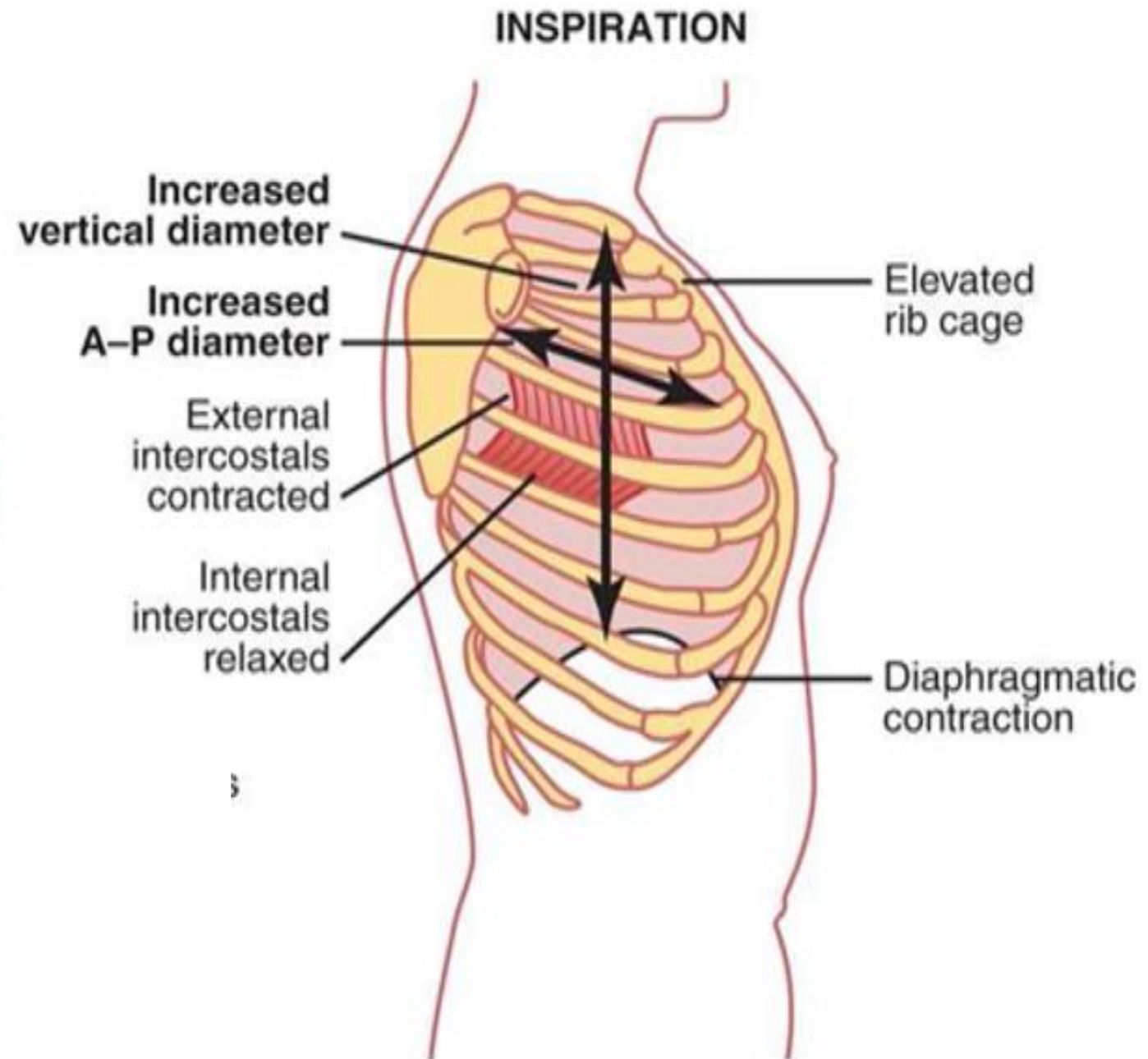
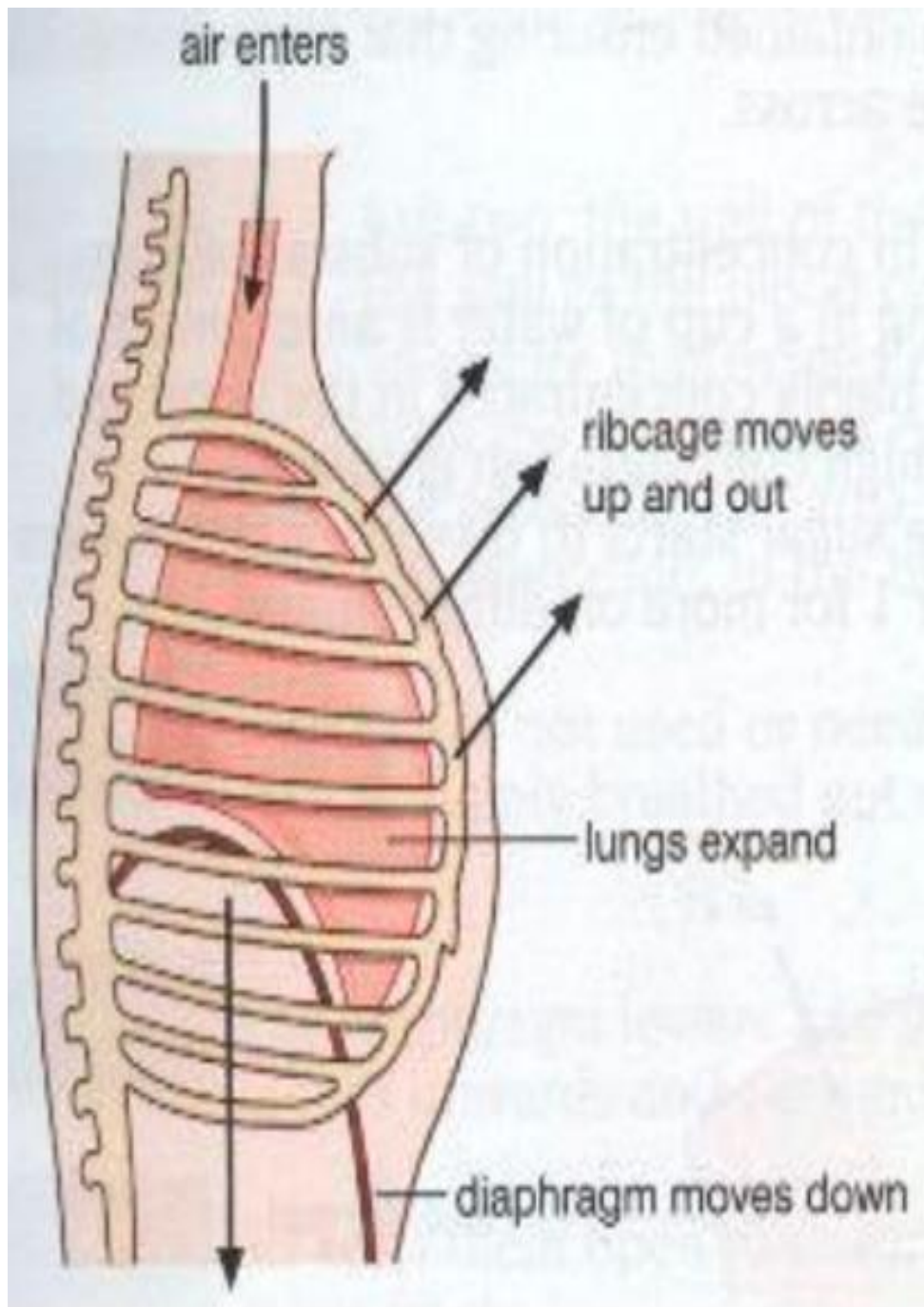
– The inflow of air into the lungs

2. Expiration/Exhalation

– The outflow of air (respiratory gases) from the lungs

Inspiration/Inhalation

- ❑ The **External Intercostal Muscles** plus the **Diaphragm** contract to bring about inspiration:
- **Contraction of External Intercostal Muscles** → elevation of ribs and sternum → ↑ antero-posterior dimension of thoracic cavity → lowers air pressure in lungs → moving of air into lungs
- **Contraction of Diaphragm** → moves of the diaphragm downward → ↑ vertical dimension of thoracic cavity → lowering of air pressure in the lungs → movement of air into the lungs
 - ☒ Lungs are stretched (i.e., intrapulmonary volume increases)
 - ☒ Intrapulmonary pressure drops below atmospheric pressure (-1 mm Hg)
 - ☒ Intrapleural pressure drops below atmospheric pressure (- 6 mm Hg)
- ❑ **Air flows into the lungs** until pressures equalizes



Inspiratory muscles contract



Thoracic cavity expands



Pleural pressure becomes more negative



Transpulmonary pressure increases



Lungs inflate

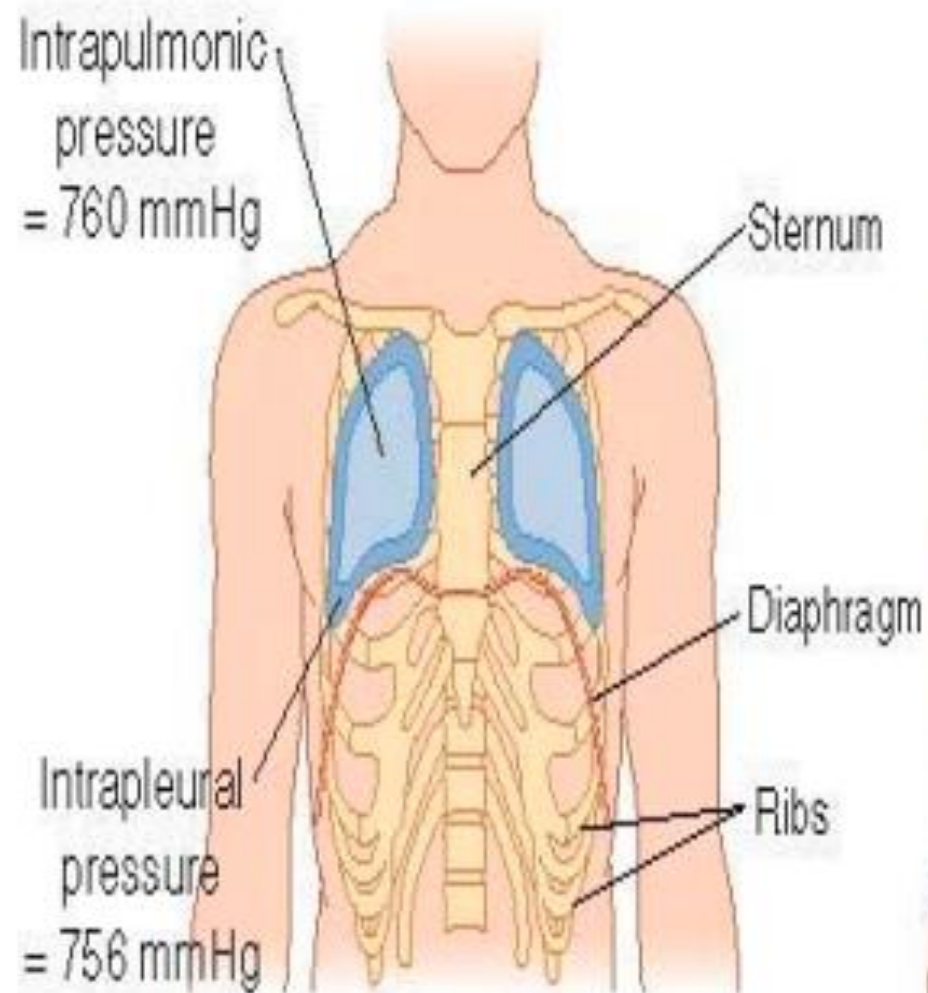


Alveolar pressure becomes subatmospheric

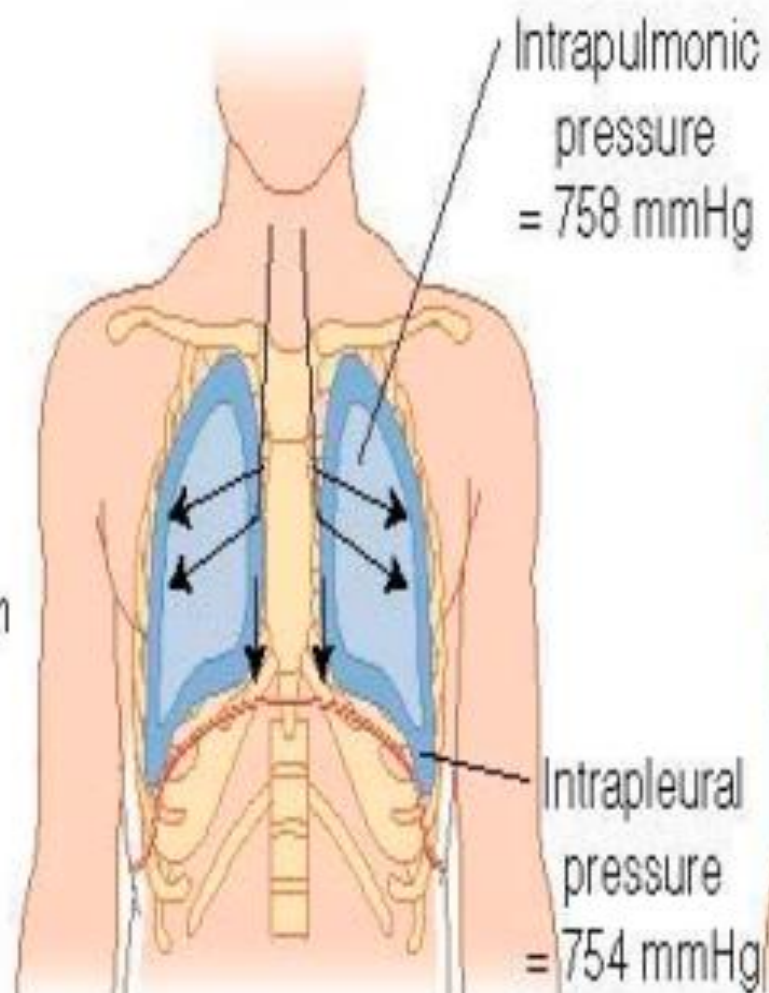


Air flows into the lungs until alveolar pressure equals atmospheric pressure

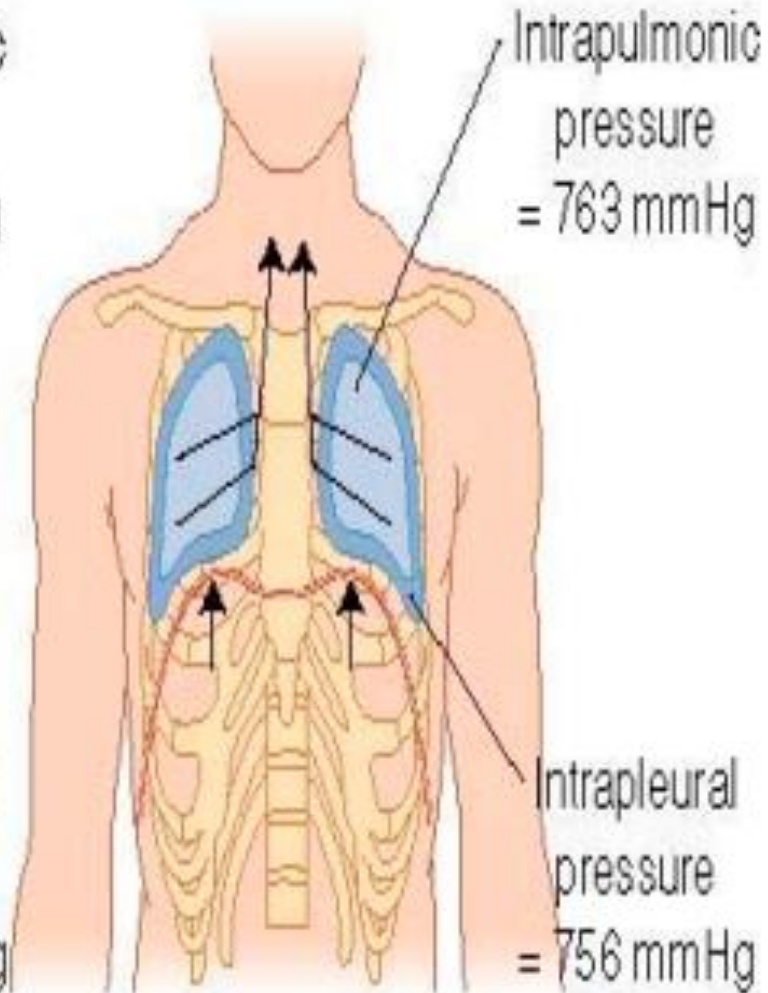
Atmospheric pressure =
760 mmHg



a. Rest



b. Inspiration



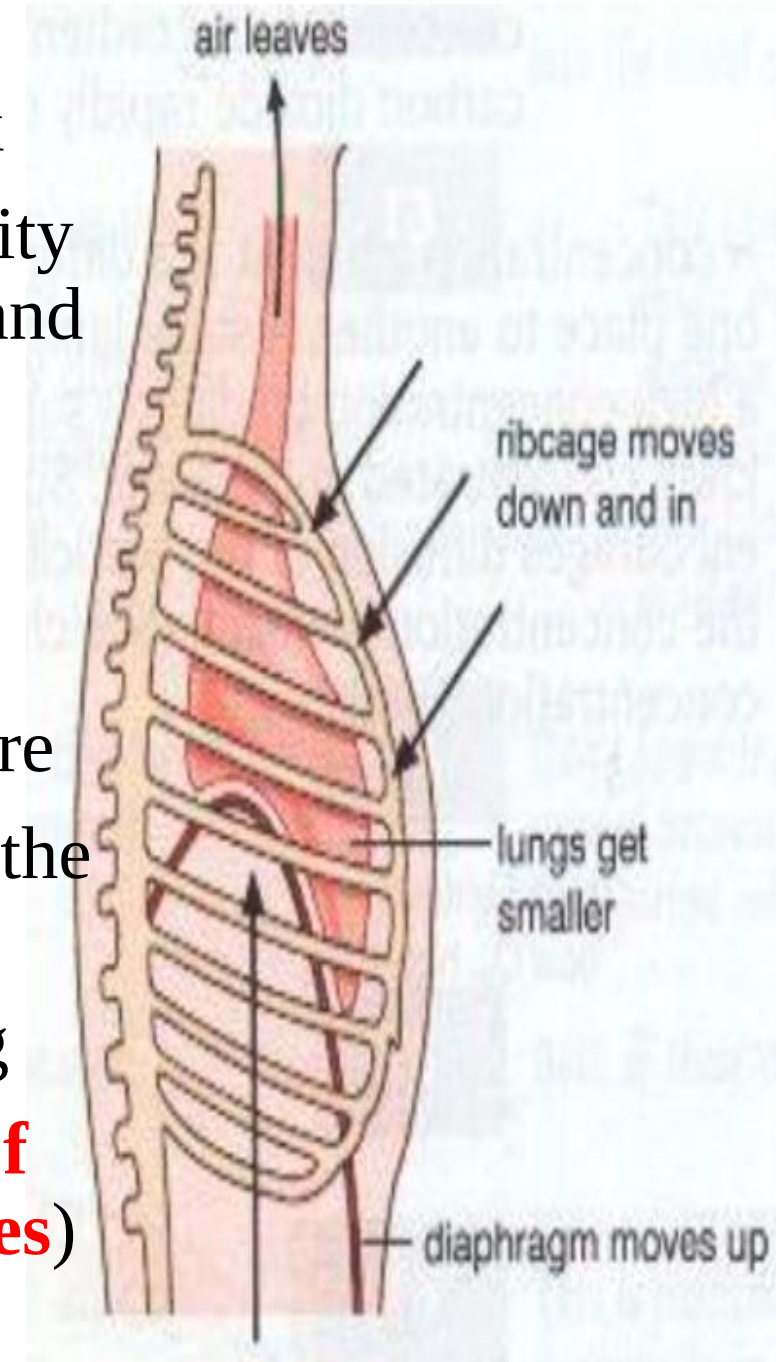
c. Expiration

Expiration / Exhalation

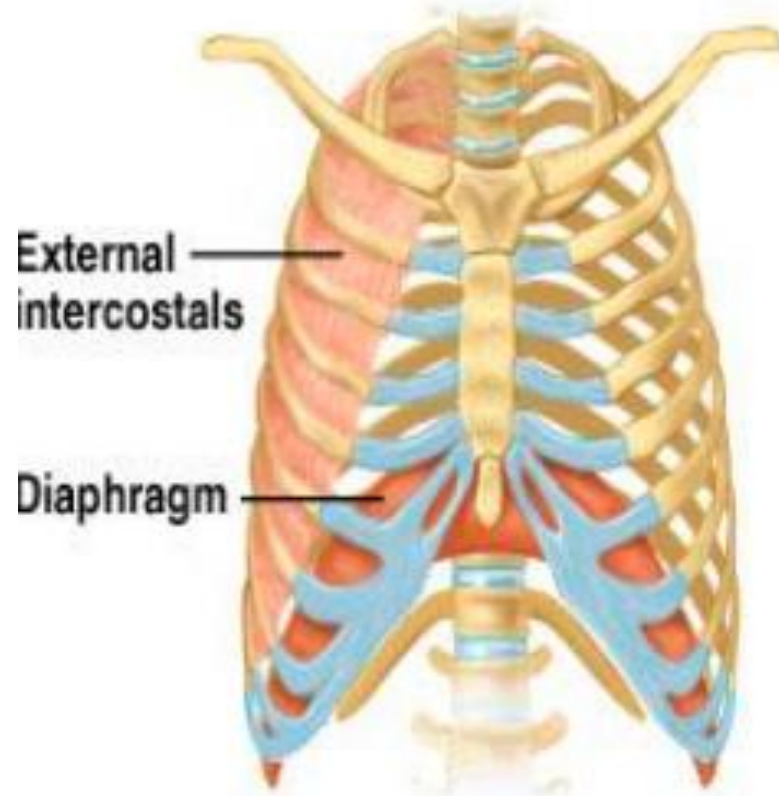
■ **Diaphragm** and **External Intercostal Muscles** relax

⇒ **Ribcage** moves downward & inward due to gravity and **diaphragm** moves upward ⇒ ↓ horizontal and vertical dimension of thoracic cavity

- ☒ Volume of thoracic cavity decreases
- ☒ Lungs recoil passively ⇒ ↓ volume of lungs
- ☒ Intrapulmonary pressure ↑ above atmospheric pressure
 - ☐ Air flows out of the lungs to the atmosphere until the two pressures equilibrate
- ☐ Expiration is a passive process during quiet breathing
- ☐ Forced expiration is active (e.g., needs **contraction of abdominal muscles and internal intercostal muscles**)

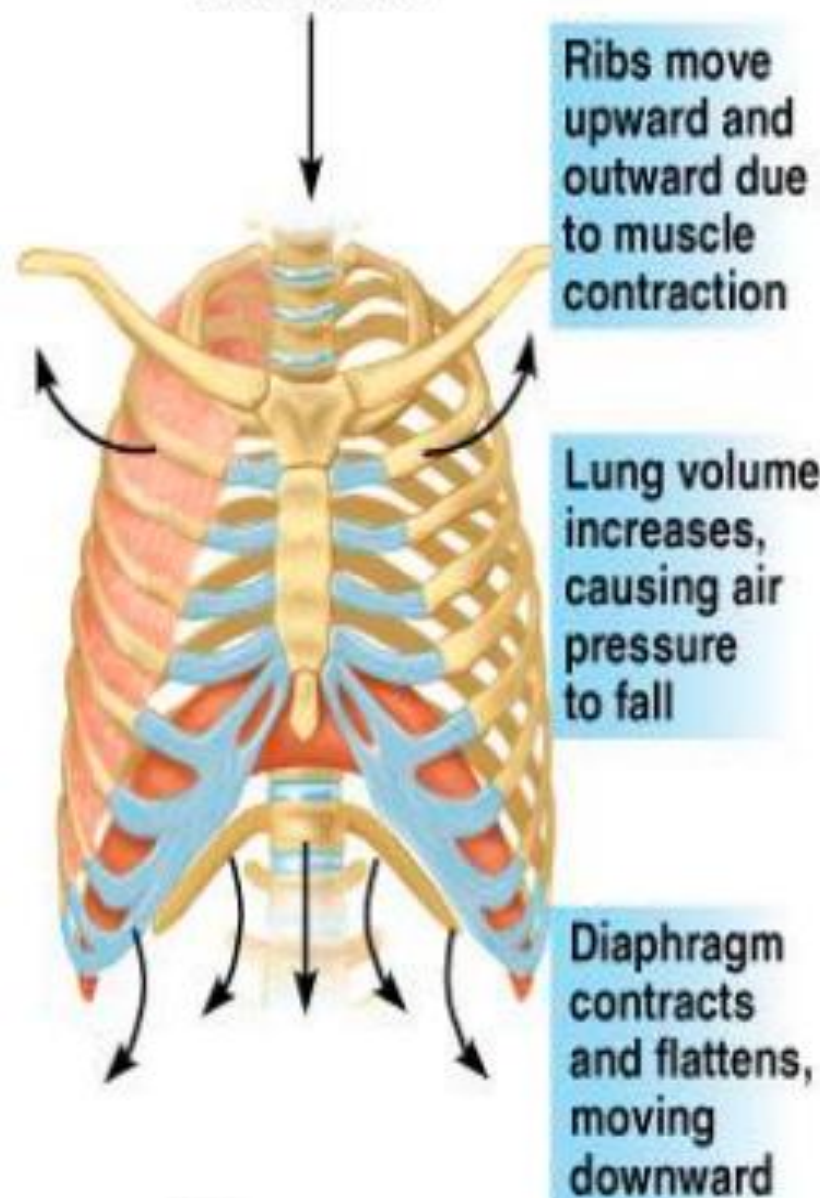


No air movement



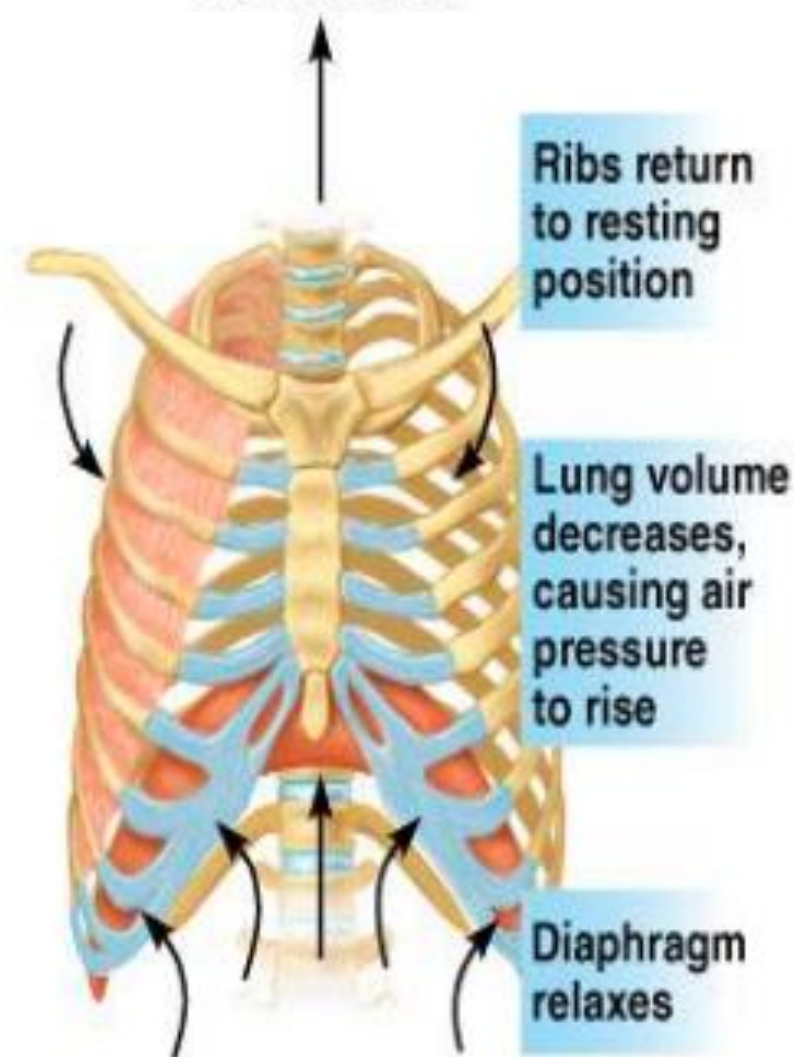
① Relaxed state

Air flows in

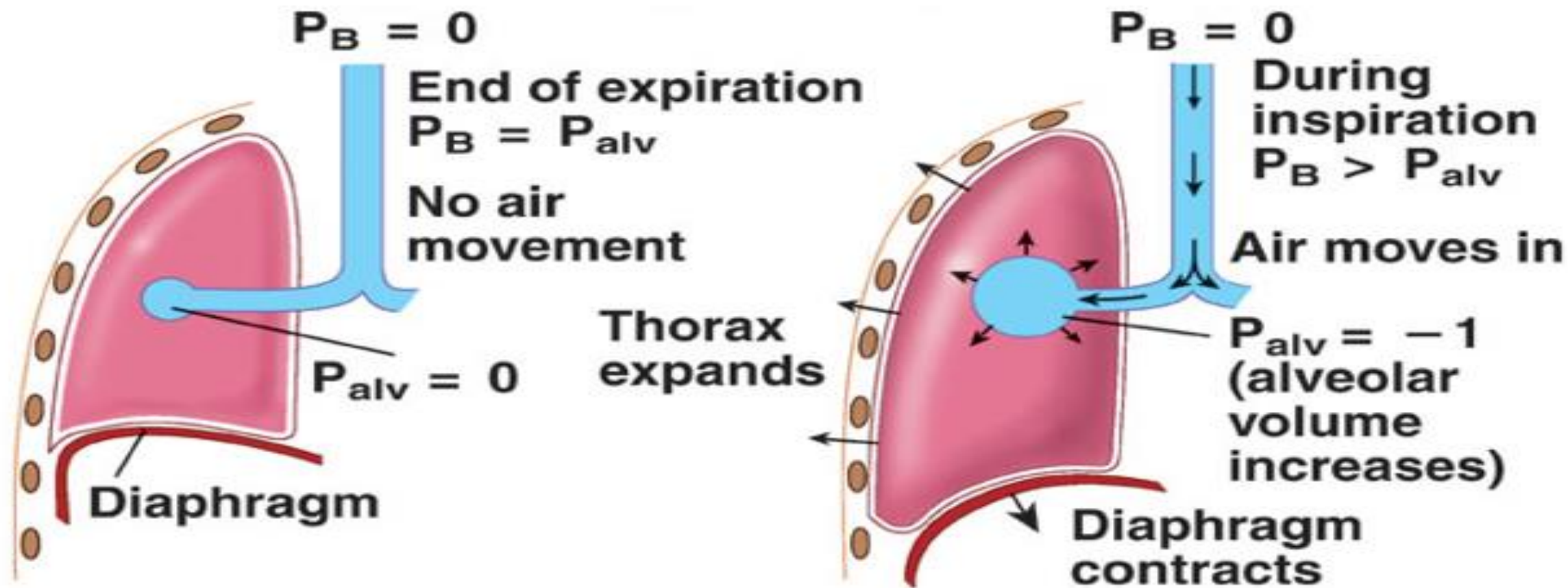


② Inspiration

Air flows out

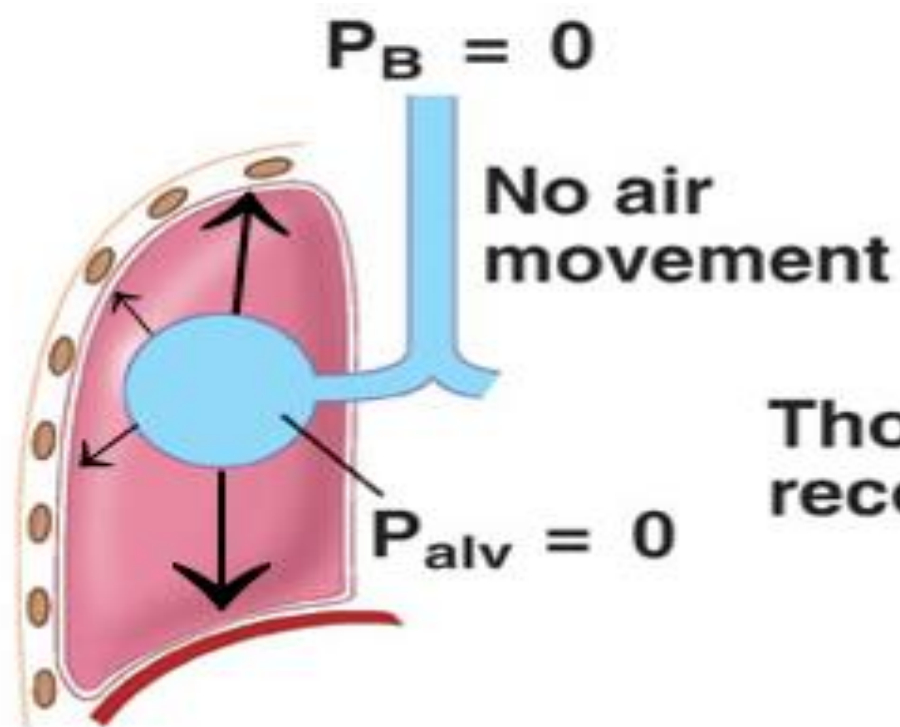


③ Expiration

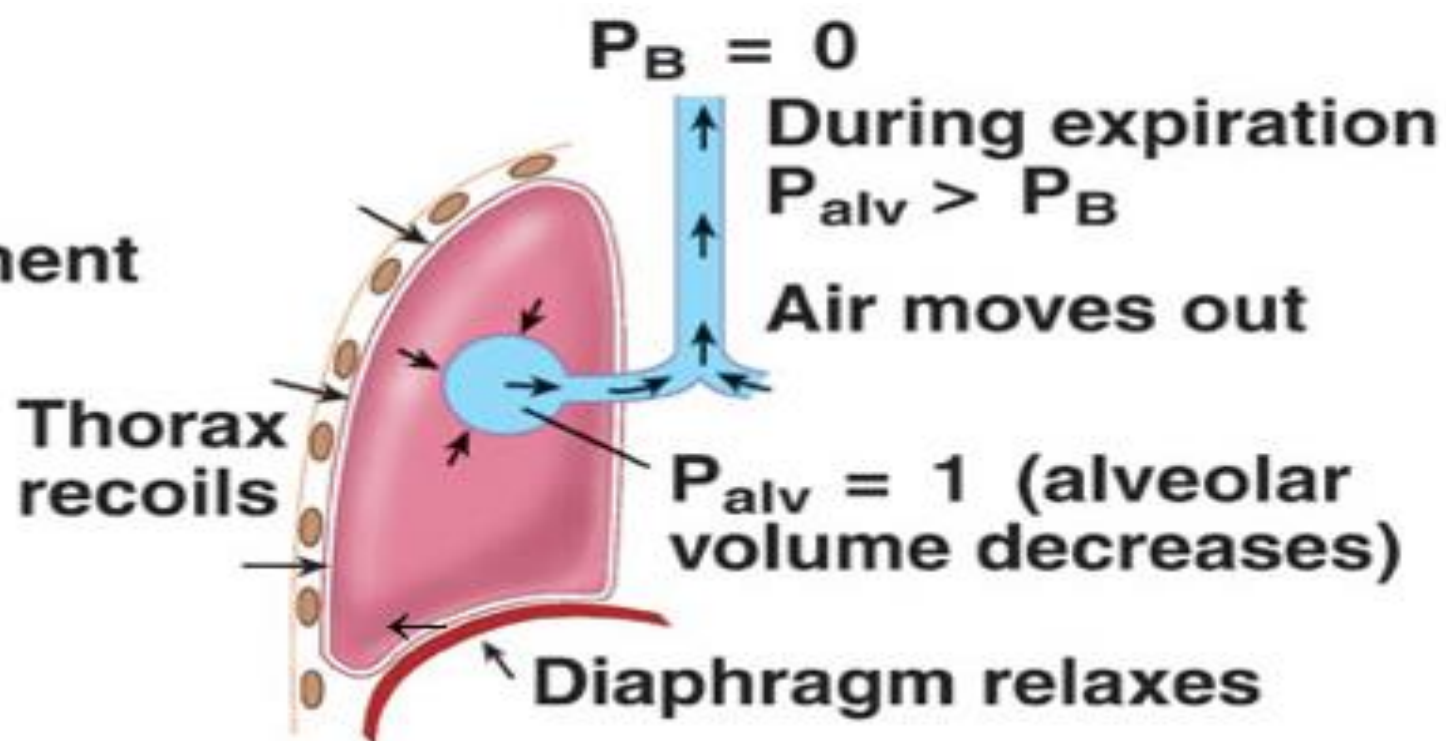


1. Barometric air pressure (P_B) is equal to alveolar pressure (P_{alv}) and there is no air movement.

2. Increased thoracic volume results in increased alveolar volume and decreased alveolar pressure. Barometric air pressure is greater than alveolar pressure, and air moves into the lungs.



3. End of inspiration.



4. Decreased thoracic volume results in decreased alveolar volume and increased alveolar pressure. Alveolar pressure is greater than barometric air pressure, and air moves out of the lungs.

Pressures of Pulmonary Ventilation

- Three pressures are important for ventilation

1. Atmospheric (barometric) Pressure

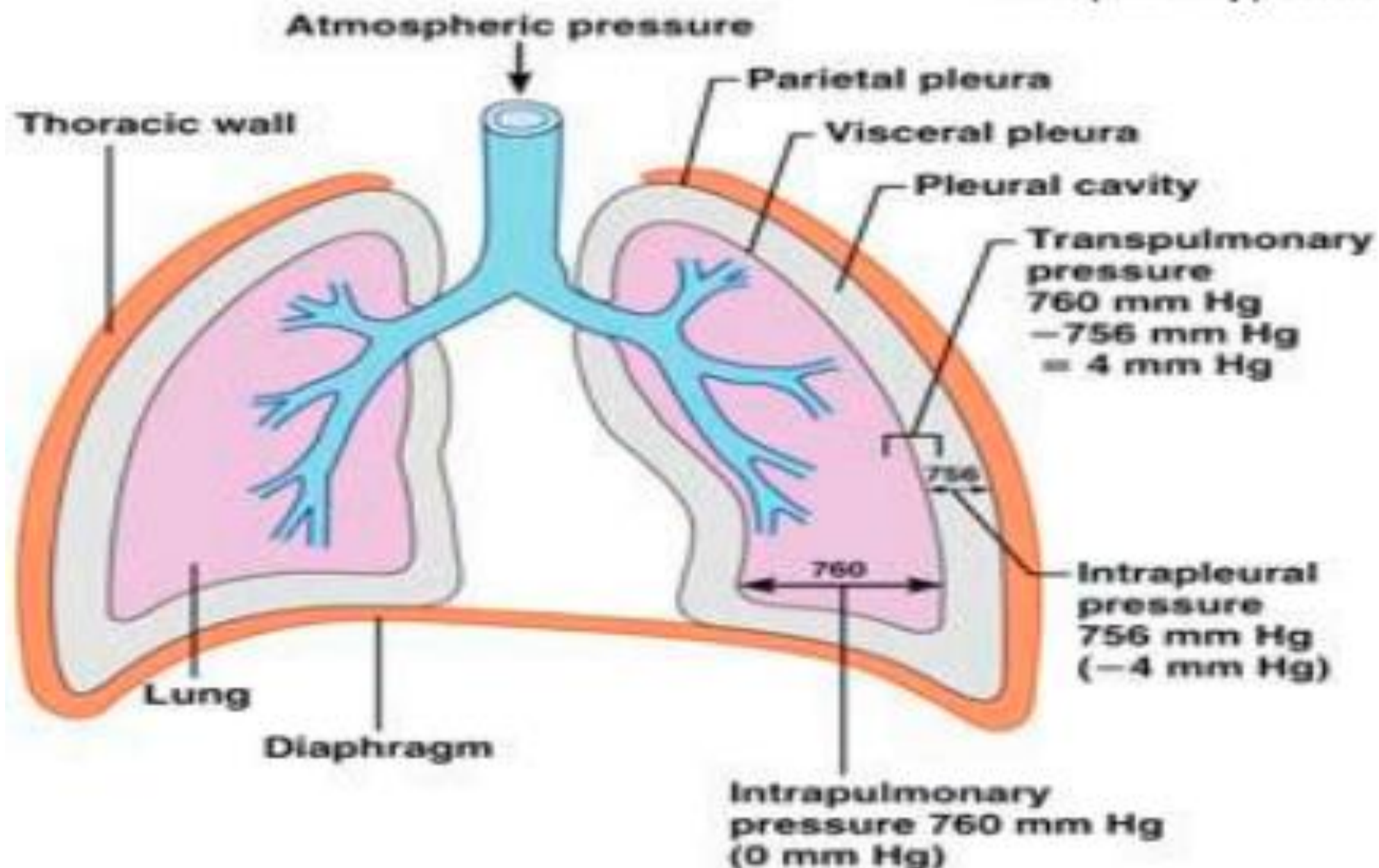
⇒ Pressure exerted by the **weight of air** in the atmosphere on **objects of Earth's surface**. At sea level, it equals **760 mm Hg**.

- Atmospheric pressure diminishes with increasing altitude above sea level as the layer of air above Earth's surface correspondingly decreases in thickness.

2. Alveolar pressure (intrapulmonary pressure)

⇒ Pressure within the alveoli

- Because the **alveoli communicate** with the **atmosphere** through the **conducting airways**, air quickly flows down its pressure gradient any time **intra-alveolar pressure differs from atmospheric pressure**; **air flow continues until the two pressures equilibrate (become equal)**.



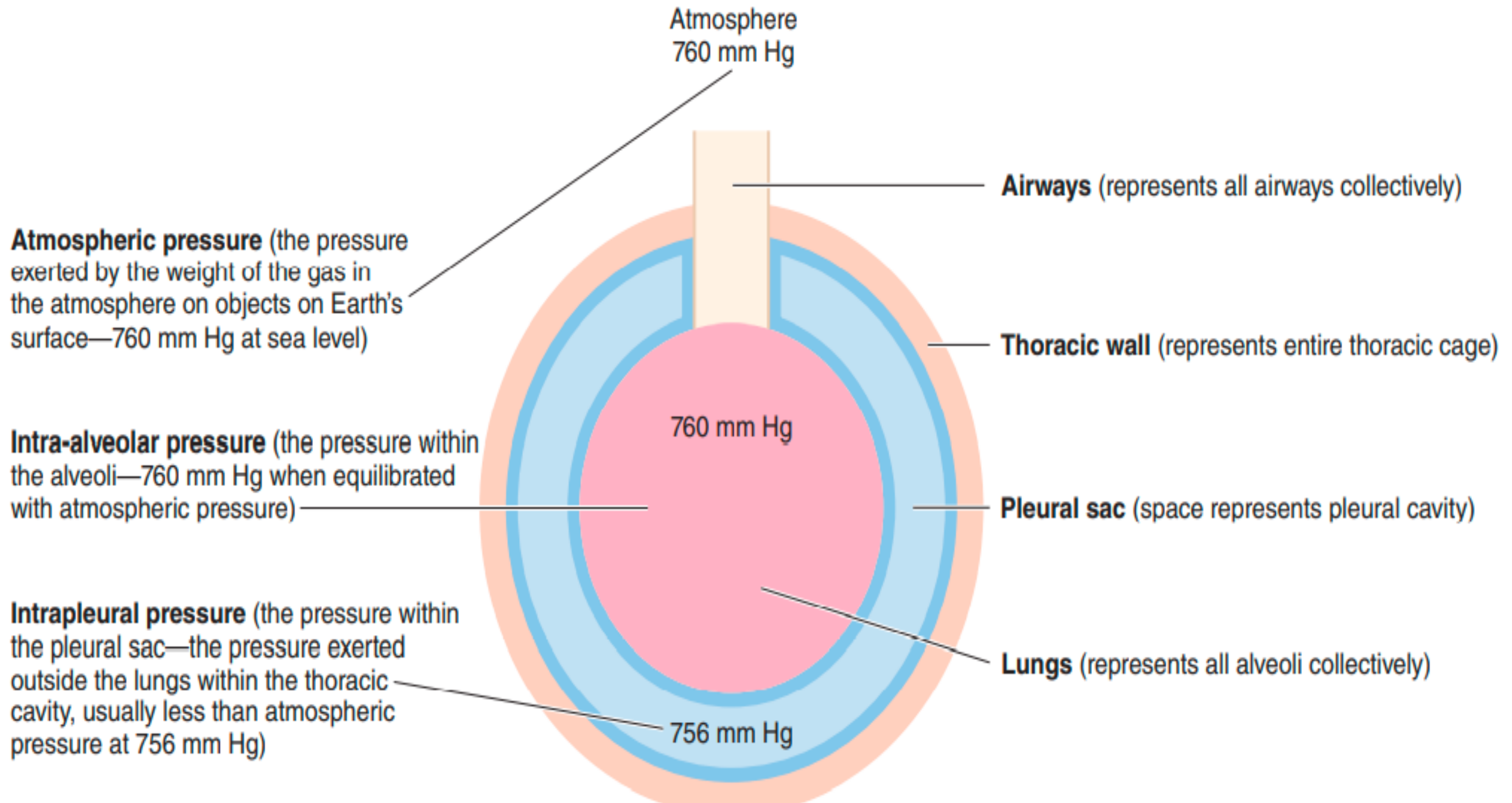
3. Pleural Pressure (Intrapleural Pressure)

⇒ Pressure within the pleural sac (i.e., It is the **pressure exerted outside the lungs** within the thoracic cavity).

□ Intrapleural pressure is usually less than atmospheric pressure, **averaging 756 mm Hg at rest.**

⇒ 756 mm Hg is sometimes referred to as a **pressure of -4 mm Hg**. A pressure of -4 mm Hg is just negative when compared with the normal atmospheric pressure of 760 mm Hg.

□ **Intrapleural pressure** does not equilibrate with atmospheric or intra-alveolar pressure **because** the **pleural sac is a closed sac with no openings**, so air cannot enter or leave despite any pressure gradients that might exist between the pleural cavity and the atmosphere or lungs.



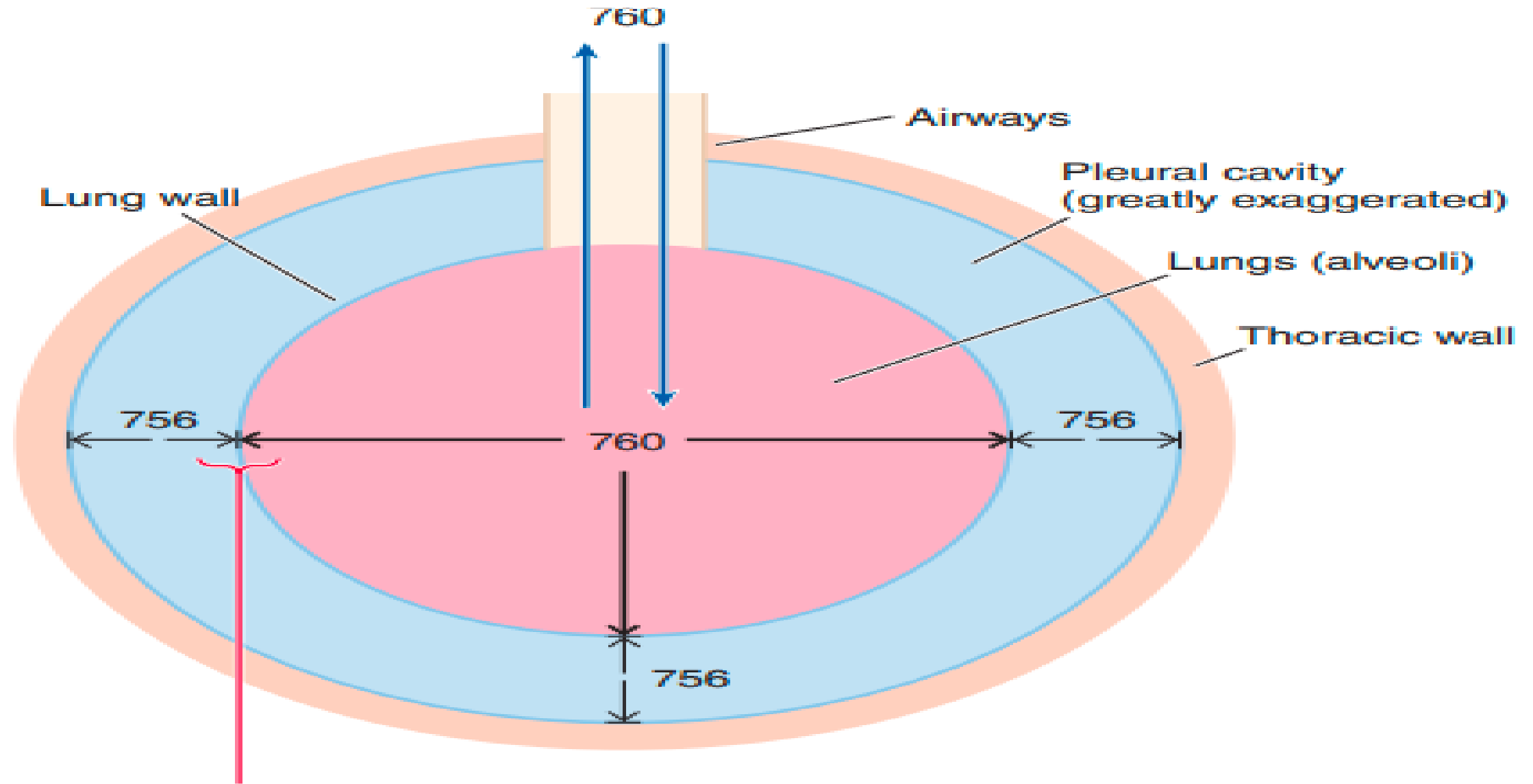
■ Transmural Pressure (Transpulmonary Pressure)

□ Intra-alveolar pressure, equilibrated with atmospheric pressure at **760 mm Hg**, is greater than the **intrapleural pressure of 756 mm Hg**, so a greater pressure is **pushing outward** than **pushing inward** across the lung wall.

⇓ This net **outward pressure difference**, the **Transmural pressure (+ 4 mm Hg)** ⇒ pushes the lung outward, causes stretching or distending.

□ Because of the **Transmural Pressure**, the **lungs are always forced to expand and to fill the thoracic cavity**, no matter its size.

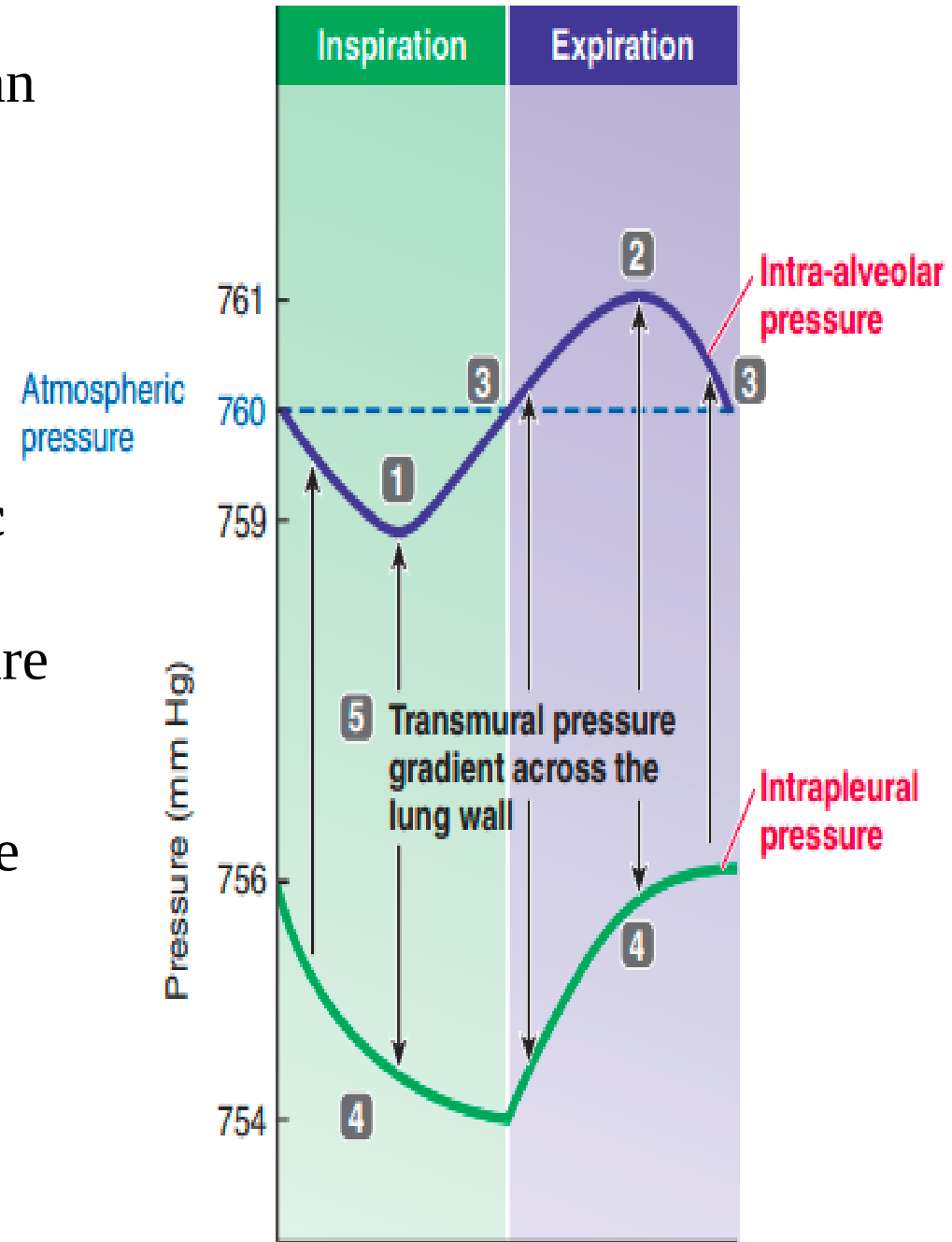
☒ As the thoracic cavity enlarges, the lungs enlarge; that is, the **lungs follow the movements of the chest wall**.



Transmurial pressure gradient
across lung wall =
intra-alveolar pressure –
intrapleural pressure

Change of Pressure during Respiration

1. **During inspiration**, intra-alveolar pressure is less than atmospheric pressure.
2. **During expiration**, intra-alveolar pressure is greater than atmospheric pressure.
3. At the end of both inspiration and expiration, intra-alveolar pressure is **equal** to atmospheric pressure **b/c** the alveoli are in direct communication with the atmosphere, and air continues to flow down its pressure gradient until the two pressures equilibrate.
4. Throughout the respiratory cycle, intrapleural pressure is less than intra-alveolar pressure.
5. Transmural pressure always exists, and the lung is always stretched to some degree, even during expiration.

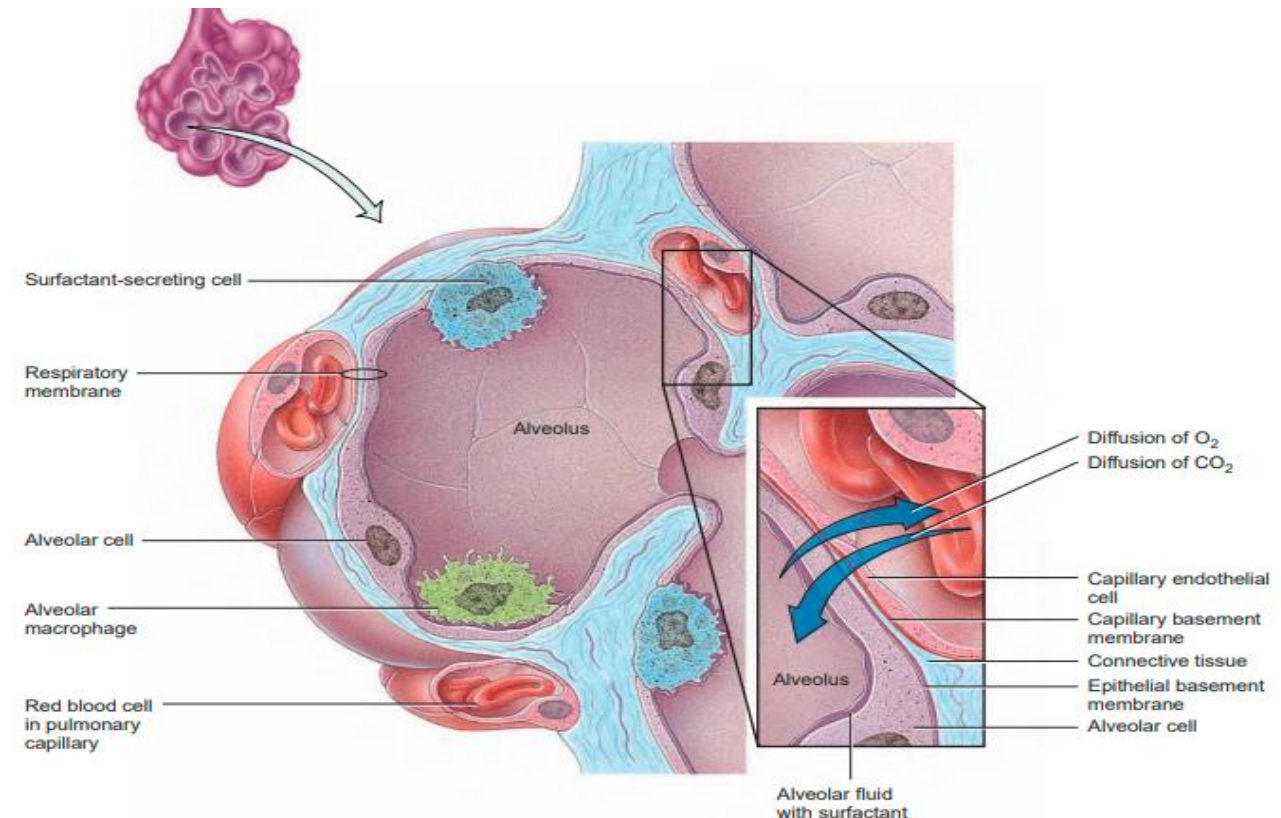


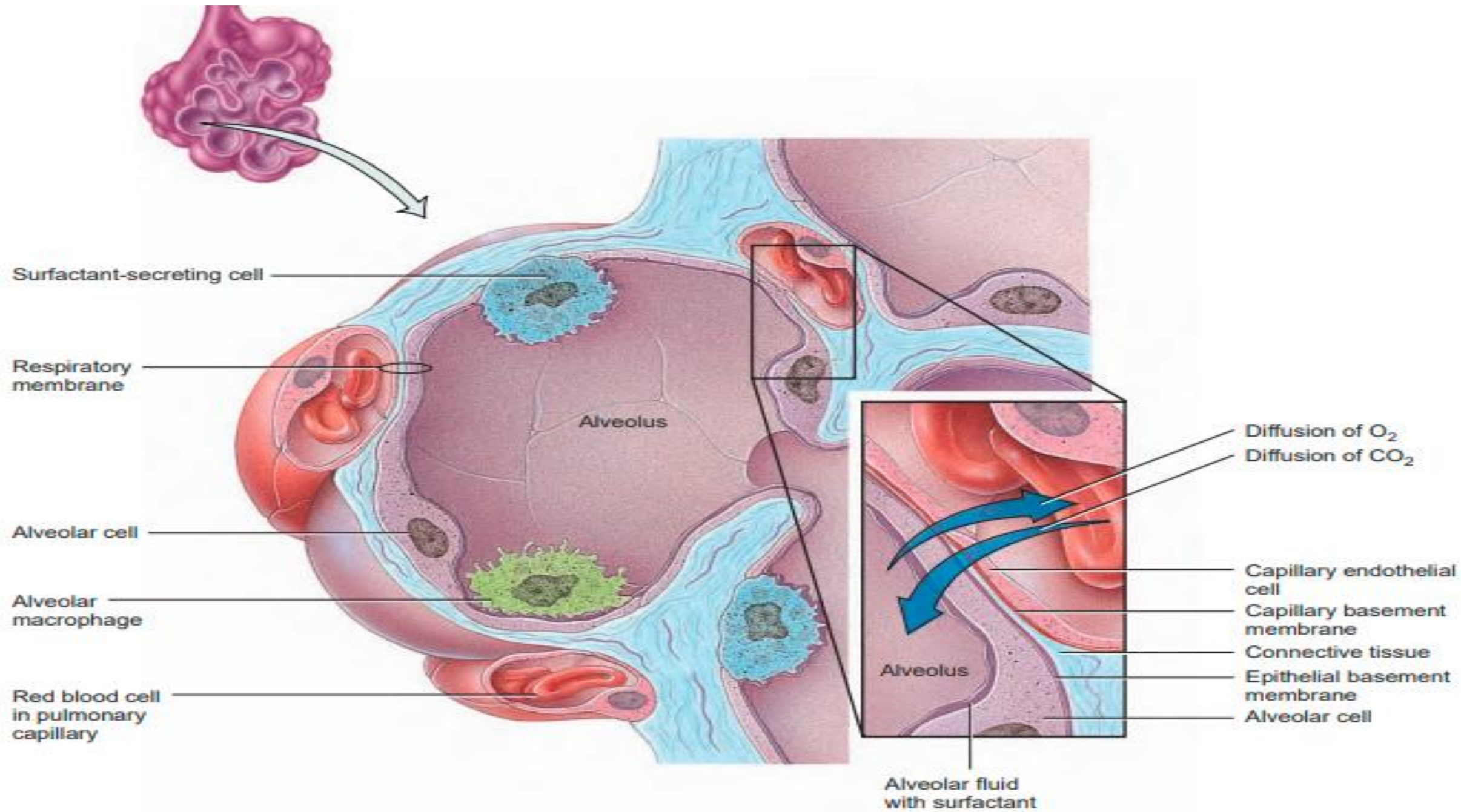
Surfactant

- ❑ **Type II alveolar cells** secrete alveolar fluid, which keeps the surface moist between **alveoli cells** and **air in the alveoli**
 - ⇒ In the alveolar fluid there is **surfactant**, a mixture of phospholipids and lipoproteins
- ☒ **Surfactant** reduces the **surface tension of alveolar fluid**, which lowers the tendency of alveoli to collapse

Actions of surfactant

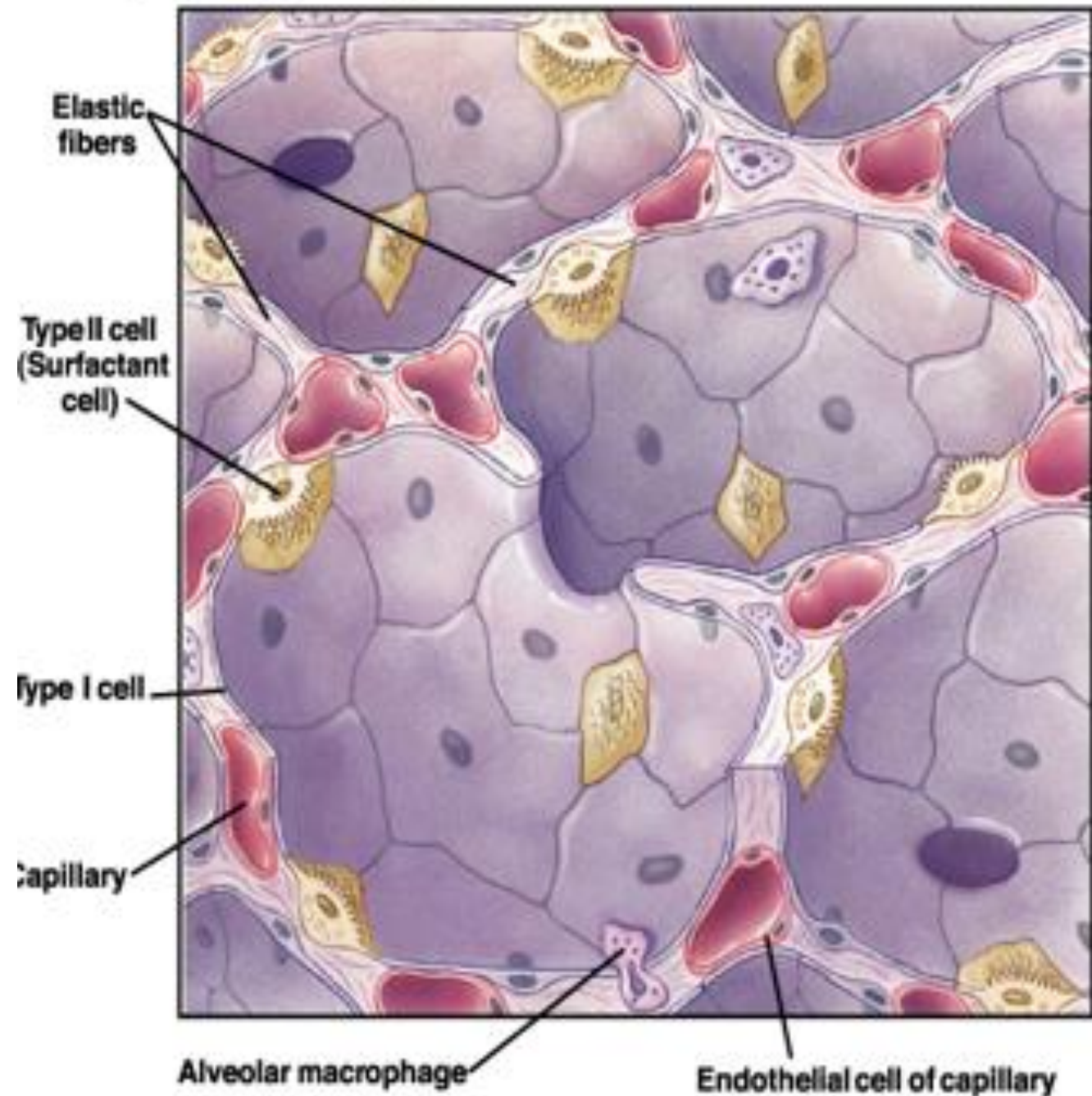
1. Distensibility of lungs
2. Reduced recoil force of lungs
3. Prevent alveolar collapse by balancing pressure b/n small & large alveoli
4. Reduce the work of breathing





Surface Tension

- ❑ Force generated in the **surface of liquid** at **gas–liquid interface**
- ❑ Occurs between the **alveolar gas** and the **fluid lining the alveoli**
- Surface tension **tends to shrink** the surface area of the interface, so the tendency for the alveolar surface area to shrink tends to decrease alveolar Volume



Role of Surfactant during Respiration

✦ **Deep inspiration** → Expansion of alveoli → ↑ surface area of fluid → dispersion of surfactant → ↓ surfactant effect → ↑ surface tension

✦ **Expiration** → recoil of alveoli to smaller size → ↓ fluid surface area → condensed surfactant → ↑ surfactant activity → ↓ surface tension

✦ **↑ Surface tension** during inspiration → prevention of alveolar over distension and rupture.

✦ **↓ Surface tension** during expiration → prevention of lung collapse.

Pulmonary Volumes and Capacities

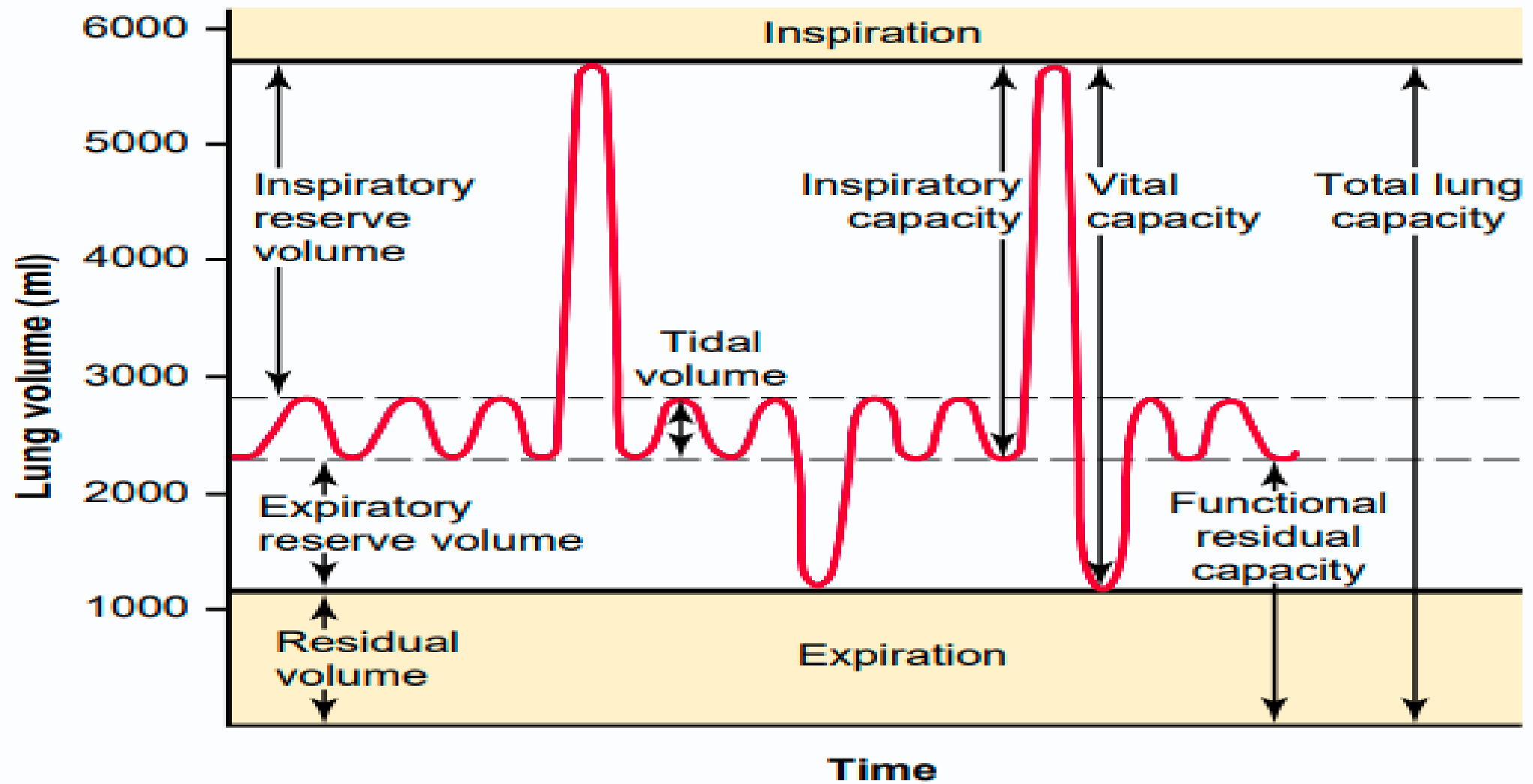
■ Spirometry

- ❑ Process of measuring **volumes of air that move into and out of** the respiratory system

■ Spirometer

- ❑ A device used to measure the pulmonary volumes
- ❑ Four **pulmonary volumes** and representative values

1. **Tidal Volume (TV)** : Volume of air inspired or expired during a single breath. At rest, quiet breathing results in a tidal volume of approximately 500 mL.
2. **Inspiratory Reserve Volume (IRV)**: Amount of air that can be **inspired forcefully after inspiration of the tidal volume** (approximately 3000 mL at rest).



3. Expiratory Reserve Volume (ERV) : Amount of air that can be **forcefully expired after** expiration of the tidal volume (approximately 1100 mL at rest).

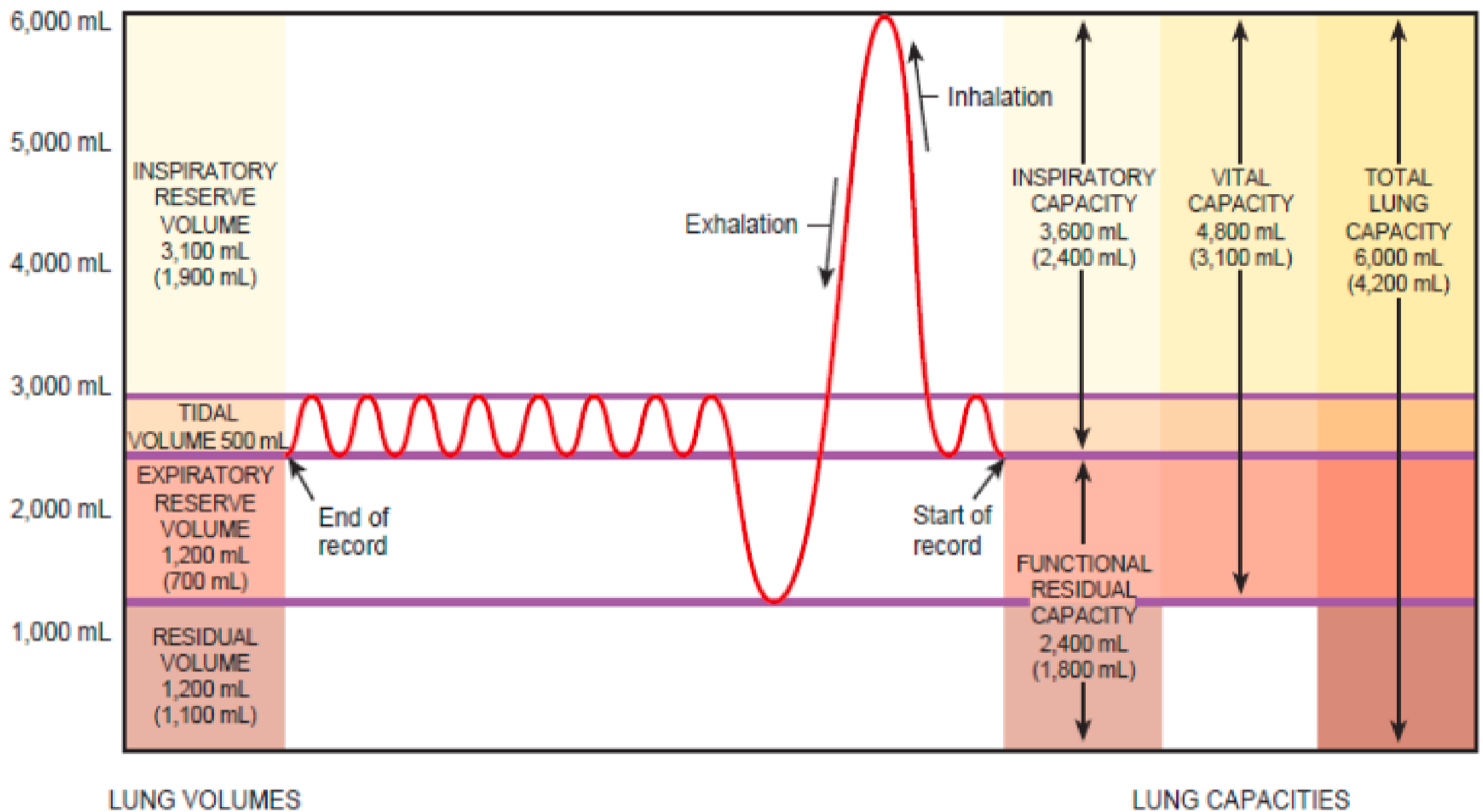
4. Residual Volume (RV) : Volume of air **remaining** in the respiratory passages and lungs **after forceful expiration** (approximately 1200 mL).

❑ **Tidal volume** increases when a person is more active. **An increase in tidal volume causes a decrease in the inspiratory and expiratory reserve volumes**

⇒ Because the maximum volume of the respiratory system does not change from moment to moment.

❑ **Pulmonary Capacities** are the sum of two or more pulmonary volumes

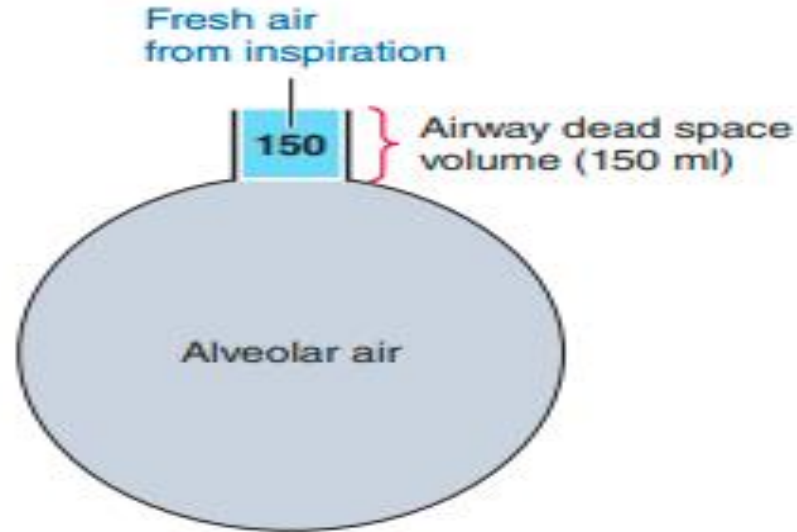
1. **Inspiratory Capacity (IC)**: The amount of air that can **inspire maximally** after a normal **expiration** (approximately 3500 mL at rest) $\Rightarrow$ $(IC = IRV + TV)$.
2. **Functional Residual Capacity (FRC)** : The amount of air **remaining in the lungs at the end of a normal expiration**. $(FRC = ERV + RV) \Rightarrow$ Average value = 2200 ml at rest
3. **Vital Capacity (VC)** : The maximum volume of air that can be **moved out during a single breath following a maximal inspiration**. The subject first inspires maximally and then expires maximally (approximately 4600 mL) $\Rightarrow$ $(VC = IRV + TV + ERV)$.
 $\Rightarrow$ valuable in determining the functional capacity of the lungs.
4. **Total Lung capacity (TLC)** : The maximum volume of air that the lungs can hold $(TLC = VC + RV) \Rightarrow$ Average value = 5700 ml.



❑ **Residual Volume; Functional Residual Capacity; and Total Lung Capacity** cannot be measured **directly with a spirometer**, because the residual volume of air does not move into and out of the lungs. It can be determined indirectly, however, through **gas-dilution techniques** involving inspiration of a known quantity of a harmless tracer gas such as helium

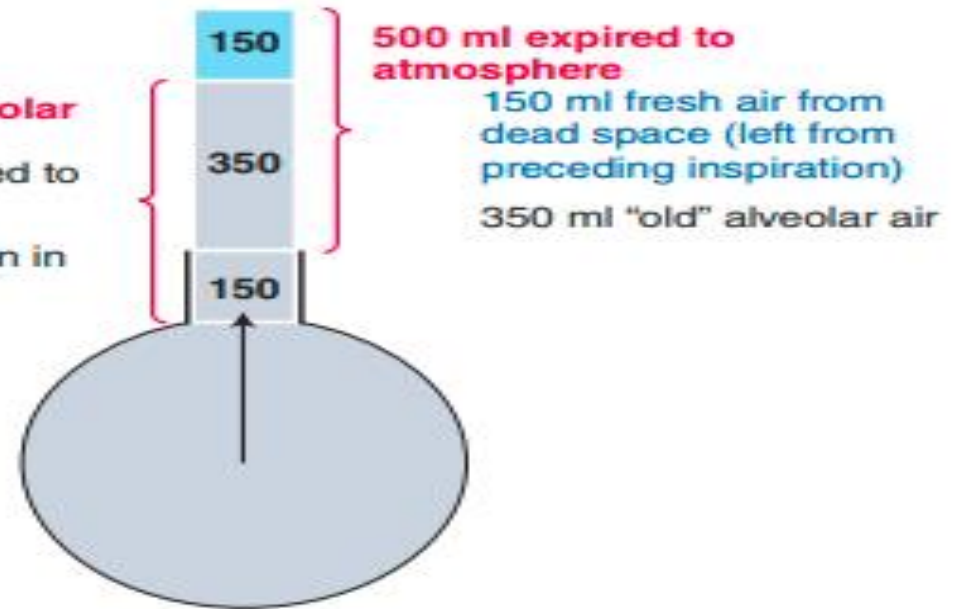
Dead Space

- ❑ Part of the respiratory system where **gas exchange does not take place**
- ❑ The nasal cavity, pharynx, larynx, trachea, bronchi, bronchioles, and terminal bronchioles form **150 mL of dead space**
- ❑ Out of **500 ml** of air that are moved into and out of the lungs with each breath, **only 350 ml are actually exchanged between the atmosphere and the alveoli** because of the 150 ml occupying the **Dead Space**.
- ❑ During inspiration, much of the inspired air fills the dead space first before reaching the alveoli and, thus, is unavailable for gas exchange.



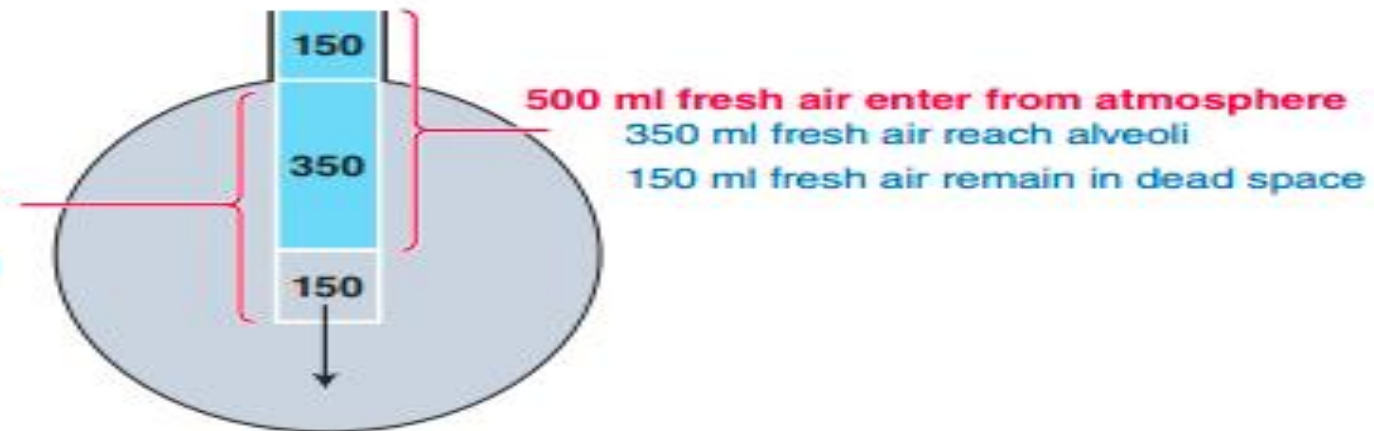
(a) After inspiration, before expiration

500 ml "old" alveolar air expired
 350 ml expired to atmosphere
 150 ml remain in dead space



(b) During expiration

500 ml enter alveoli
 150 ml "old" air from dead space (left from preceding expiration)
 350 ml fresh air from atmosphere



(c) During inspiration

■ **Total Dead Space** (physiological dead space)

- ❑ Sum of alveolar and anatomical dead spaces

1. **Anatomical Dead Space**

- ❑ The air in the respiratory system that extends from **nasal cavity** to **terminal bronchioles**
- ❑ The **air does not contact** with the **alveoli** of the lungs
- ❑ Gas exchange does not take place

2. **Alveolar Dead Space**

- ❑ extends from **respiratory bronchioles** to **alveolar ducts**
- ❑ The **air contacting** with **alveoli**
- ❑ Gas exchange take place **with out** contacting **blood flow** in the pulmonary capillaries (i.e. ventilation without perfusion)

☒ Alveolar ventilation is less than pulmonary ventilation because of **dead space**

☐ **Pulmonary Ventilation (Minute Ventilation)**

☒ The total amount of air moved into and out of the respiratory system each minute

☒ It is equal to **tidal volume** times **respiratory rate**.

■ **Respiratory rate** is the number of breaths taken per minute

☐ **Pulmonary Ventilation** on averages is approximately **6 L/min** because the tidal volume is 500 mL and respiratory rate is 12 breaths per minute.

☐ Although minute ventilation measures the amount of air moving into and out of the lungs per minute, it is not measure the amount of air available for gas exchange because gas exchange takes place mainly in the alveoli and to a lesser extent in the alveolar duct and respiratory bronchioles.

Alveolar Ventilation

- ❑ The volume of air exchanged between the **atmosphere** and **alveoli** per minute
- ❑ The amount of atmospheric air that reaches the alveoli and is actually available for exchange with blood is more important than the total amount breathed in and out
- ❑ In determining alveolar ventilation, the amount of wasted air moved in and out through the anatomic dead space must be taken into account

Alveolar ventilation = (tidal volume – dead space volume) × respiratory rate

$$\begin{aligned}\text{Alveolar ventilation} &= (500 \text{ ml/breath} - 150 \text{ ml}) \times 12 \text{ breaths/min} \\ &= \mathbf{4200 \text{ ml/min}}\end{aligned}$$

▲ Table 12-2 Effect of Different Breathing Patterns on Alveolar Ventilation

Breathing Pattern	Tidal Volume (ml/breath)	Respiratory Rate (breaths/min)	Dead Space Volume (ml)	Pulmonary Ventilation (ml/min)*	Alveolar Ventilation (ml/min)**
Quiet breathing at rest	500	12	150	6000	4200
Deep, slow breathing	1200	5	150	6000	5250
Shallow, rapid breathing	150	40	150	6000	0

*Equals tidal volume $\times$ respiratory rate.

**Equals (tidal volume – dead space volume) $\times$ respiratory rate.

Physical Principles of Gas Exchange and Gas Transport